IEEE
AI Team

Uber
PRICE REGRESSION
USING CLASSICAL MACHINE LEARNING SOLUTION

- PREDICTING FARE ACCURATELY
- DATA DRIVEN INSIGHTS
- CLASSICAL ML MODELS
- BETTER PREDICTIONS BETTER DECISIONS

$$\hat{y} = \beta_0 + \sum_{i=1}^{n} \beta_i x_i + \varepsilon$$

KEY FEATURES

- ✓ Distance
- ✓ Time
- ✓ Traffic Conditions
- ✓ Weather
- ✓ Location & More

DATA → DATA → MODEL → PREDICTION &

Objective

Develop a complete **Uber Price Regression System** that predicts the estimated fare of an Uber trip using **Classical Machine Learning** techniques. The project must follow a complete machine learning pipeline, starting from data preprocessing and feature engineering, through model training and evaluation, and ending with deployment as an interactive web application using **Streamlit**.

The solution should be designed as a real-world software project, emphasizing clean architecture, maintainability, scalability, and software engineering best practices.

Project Requirements

Data Preprocessing

- Handle missing values appropriately.
- Remove duplicated records.
- Detect and treat outliers where necessary.
- Perform data cleaning before model training.
- Apply suitable encoding techniques for categorical features.
- Normalize or standardize numerical features whenever appropriate.

Feature Engineering

You must create meaningful new features from the original dataset. Examples include (but are not limited to):

- Trip distance.
- Time-based features (hour, day, month, weekend, rush hour).
- Direction or bearing between pickup and drop-off.
- Geographical features.
- Distance from city center.
- Any additional engineered features that improve prediction performance.

you are encouraged to justify every engineered feature and explain its expected impact on the prediction model.

Feature Selection

Apply one or more feature selection techniques to identify the most informative features. Possible techniques include:

- Correlation Analysis
- Mutual Information
- Recursive Feature Elimination (RFE)
- Feature Importance
- Permutation Importance
- SHAP-based importance (optional)

The final selected feature set should be clearly documented.

Machine Learning Models

Train and compare multiple **Classical Machine Learning Regression** algorithms such as:

- Linear Regression
- Ridge Regression
- Lasso Regression
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting Regressor

you should compare model performance and select the best-performing model.

Model Evaluation

Evaluate the trained models using appropriate regression metrics, including:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Visualization of prediction results is highly encouraged.

Streamlit Deployment

Develop an interactive **Streamlit** application that allows users to estimate Uber trip prices.

The application must include:

- User-friendly interface.
- Interactive map.
- Pickup location selection directly from the map.
- Drop-off location selection directly from the map.
- Automatic calculation of trip-related features.
- Fare prediction using the trained machine learning model.
- Display of predicted price and trip information.

SQLite Database Integration

Integrate an **SQLite3** database to store every prediction request.

Each prediction record should include information such as:

- Pickup latitude and longitude
- Drop-off latitude and longitude
- Date and time
- Engineered features (if desired)
- Predicted fare
- Timestamp of prediction

The application should automatically save every user request into the database.

Software Engineering Requirements

The project **must not** be implemented in a single file.

Students are required to organize the project into multiple modules following **Object-Oriented Programming (OOP)** principles and the **SOLID Design Principles**.

ركزوا معايا هنا, البروجيكت ده صعب وانا متعمد انه صعب علشان اسباب معينه هتعرفوها بعدين بعد البروجيكت, المهم انه في جزء self learning كتير وحجات انا مشرحتهاش ولكن هنا هتكون الشطاره, ياتري مين التيم اللي هيشتغل علي نفسه ويذاكروا علشان يعملوا البروجيكت ده

عموما يعني مجال ال AI من اكتر المجالات اللي فيها ريسيرش, والبروجيكت ده هينمي الحتة ديه عندكوا جدا لان اكتر من نصه عباره عن ريسيرش حرفيا بطريقه بحتة,

المهم ان هيحصل مناقشة في الاخر اوفلاين للتيم واللي بره اسكندرية هتكون مناقشة اونلاين, فضروري الالتزام لاني هسال في كل تفصيله علشان اتاكد هل عملتوها بال AI ولا بالمجهود الشخصي

ابحثوا علي اليوتيوب علي الحجات الموجودة زي ال OOP و زي Feature engineering وعن موديلز Regression غير اللي احنا شرحناها

بالتوفيق باذن الله تعالي